

DAVIS STATION, Antarctica

WMO: 895710

Lat: 68.574S

Lon: 77.967E

Elev: 76

StdP: 14.66

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-25.0	-21.1	-34.2	0.8	-23.9	-30.5	1.0	-19.4	60.6	19.5	55.0	19.2	6.6	180	1.197

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	5.8	41.2	33.1	39.2	32.0	37.5	31.2	34.0	39.3	32.9	37.7	31.9	36.4	17.7	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
30.1	24.3	33.3	28.7	22.7	33.2	27.2	21.2	32.9	12.7	39.6	12.2	37.9	11.8	36.6	39.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 49.9	42.7	35.2	DB	-27.2	45.2	6.9	2.9	-32.1	47.3	-36.2	49.0	-40.0	50.6	-45.0	52.7	(4)
(5)			WB	-27.5	36.0	6.8	1.7	-32.4	37.2	-36.4	38.2	-40.2	39.1	-45.1	40.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	13.7	33.4	27.8	17.3	7.7	3.4	3.4	1.2	1.2	3.8	10.3	24.2	32.0	
	DBStd	14.75	3.10	4.25	7.34	9.78	10.96	10.52	12.25	10.84	9.91	8.03	5.96	3.50	
	HDD50	13236	514	622	1014	1268	1445	1398	1512	1514	1387	1232	773	557	
	HDD65	18711	979	1042	1479	1718	1910	1848	1977	1979	1837	1697	1223	1022	
	CDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	
Wind	WSAvg	13.9	14.0	14.7	13.9	12.3	13.1	13.6	14.0	13.9	13.0	13.3	15.9	14.7	
Precipitation	PrecAvg	14.2	1.5	1.5	1.3	0.9	1.1	1.1	1.2	1.1	0.9	1.0	1.4	1.3	
	PrecMax	18.6	3.7	2.7	2.7	2.2	2.3	2.5	3.1	3.5	2.3	2.1	3.2	3.0	
	PrecMin	7.7	0.6	0.8	0.4	0.3	0.4	0.2	0.1	0.1	0.1	0.3	0.4	0.3	
	PrecStd	2.2	0.7	0.5	0.6	0.5	0.5	0.5	0.6	0.7	0.5	0.4	0.7	0.6	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	45.3	40.9	33.6	30.1	30.2	27.6	28.2	27.4	26.0	29.8	38.3	44.1	
		MCWB	35.3	33.1	28.5	26.0	25.5	24.2	24.5	24.4	23.2	24.4	31.0	34.3	
	2%	DB	41.7	37.0	30.6	27.0	25.2	24.5	24.7	22.8	23.5	26.8	35.8	41.0	
		MCWB	33.4	30.7	25.9	23.5	22.2	21.6	22.1	20.0	20.8	22.7	29.5	32.7	
	5%	DB	39.7	35.2	28.6	24.2	22.4	21.6	21.4	19.8	21.0	24.5	33.8	38.9	
		MCWB	32.4	29.5	24.4	21.4	19.8	18.9	19.0	17.5	18.1	20.8	28.4	31.8	
	10%	DB	38.1	33.7	26.9	21.3	19.3	18.8	18.8	16.9	18.3	22.0	32.0	37.1	
		MCWB	31.6	28.5	23.1	18.7	16.9	16.6	16.6	14.8	15.7	18.8	27.6	31.0	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	33.7	29.2	26.4	26.0	24.8	25.4	24.6	23.9	25.2	32.2	35.4
			MCDB	43.7	38.6	32.9	29.3	29.1	27.2	27.3	27.0	25.3	28.5	35.1	41.7
2%		WB	34.3	31.9	26.7	23.9	22.4	21.8	22.2	20.2	20.9	23.0	30.5	33.8	
		MCDB	39.8	35.3	29.7	26.6	24.7	24.2	24.3	22.5	23.5	26.2	34.3	39.1	
5%		WB	33.3	30.5	24.9	21.7	20.0	19.1	19.2	17.6	18.4	21.0	29.2	32.6	
		MCDB	38.3	34.1	28.1	24.3	22.3	21.4	21.3	19.8	20.9	24.2	33.1	37.5	
10%		WB	32.3	29.2	23.2	18.7	16.9	16.6	16.5	14.9	15.7	18.8	27.9	31.6	
		MCDB	37.1	33.1	26.6	21.3	19.2	18.9	18.7	16.8	18.1	22.0	31.7	36.2	
Mean Daily Temperature Range		5% DB	MDBR	5.8	6.1	7.9	8.8	9.4	9.2	9.2	9.8	9.7	9.3	7.6	6.1
			MCDBR	7.8	7.1	7.0	7.8	9.5	9.3	7.4	10.4	9.0	9.2	8.3	7.8
	MCWBR		5.3	5.6	6.0	7.2	8.9	8.5	7.0	9.7	8.3	7.8	6.4	5.1	
	5% WB	MCDBR	6.8	6.1	6.9	8.0	9.6	9.6	6.9	10.5	9.4	9.1	7.7	6.8	
		MCWBR	5.0	5.3	6.4	7.6	9.2	9.0	6.8	10.0	8.9	7.9	6.2	4.9	
Clear-Sky Solar Irradiance	taub	0.265	0.254	0.227	0.173	0.124	N/A	0.127	0.168	0.209	0.223	0.249	0.266		
	taud	2.429	2.435	2.362	2.155	1.917	N/A	1.894	2.108	2.263	2.282	2.327	2.374		
	Ebn at Noon	314	299	273	211	50	0	37	207	275	308	318	318		
	Edh at Noon	30	26	21	11	2	0	1	12	22	30	33	34		
All-Sky Solar Radiation	RadAvg	2239	1412	728	238	23	0	5	128	580	1359	2161	2542		
	RadStd	134	112	49	14	2	0	0	10	40	56	113	140		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page